

Supplementary Materials: Algorithmic Objectivity for Congressional Redistricting

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Supplementary Material A: Edge-Weighting Technical Specification

0.1 Edge Weight Formula

For an edge (i, j) connecting adjacent census tracts i and j , the weight is computed as:

$$w(i, j) = \frac{\alpha \cdot s_{demo}(i, j)}{d(i, j) + \epsilon} \quad (1)$$

where:

- $d(i, j)$ = Euclidean distance between tract centroids (kilometers)
- $s_{demo}(i, j)$ = Demographic similarity score $\in [0, 1]$
- α = Scaling parameter (default: 10.0)
- ϵ = Small constant to prevent division by zero (default: 0.1 km)

0.2 Demographic Similarity Measure

Demographic similarity uses cosine similarity of racial/ethnic composition vectors:

$$s_{demo}(i, j) = \frac{\vec{v}_i \cdot \vec{v}_j}{|\vec{v}_i| \cdot |\vec{v}_j|} \quad (2)$$

where $\vec{v}_i = [p_{white}, p_{black}, p_{hispanic}, p_{asian}, p_{other}]$ is the 5-dimensional vector of population proportions for tract i , normalized to sum to 1.

Rationale: Cosine similarity ranges from 0 (orthogonal, maximally dissimilar) to 1 (identical proportions). This metric:

- Treats all racial/ethnic groups symmetrically
- Is insensitive to absolute population sizes (uses proportions)
- Produces intuitive results: two tracts both 70% Black have $s_{demo} \approx 1$
- Avoids requiring explicit thresholds for “minority” vs. “majority”

0.3 Distance Normalization

The denominator $d(i, j) + \epsilon$ ensures:

- Weights decay with distance (inversely proportional)
- Adjacent tracts (small d) receive higher weights than distant tracts
- The constant $\epsilon = 0.1$ km prevents infinite weights for tracts sharing long boundaries (where centroids may be very close)
- Typical tract centroid distances: 0.5-5 km within urban areas, 5-50 km in rural areas

0.4 Scaling Parameter α

The parameter α controls the relative importance of demographic similarity versus geographic compactness:

- $\alpha = 0$: Pure geographic compactness (all weights = 0, METIS minimizes geometric edge-cut)
- $\alpha = 10$ (default): Balanced weighting for VRA compliance without over-optimization
- $\alpha = 100$: Strong preference for preserving minority communities (may reduce compactness)

In our analysis, $\alpha = 10$ was selected through sensitivity testing to achieve:

- Sufficient minority-minority district creation to satisfy VRA geographic compactness requirements
- Maintained overall compactness (Polsby-Popper scores > 0.3)
- Avoided pathological cases where high weights create non-compact districts

0.5 Weight Distribution Statistics

Across all 50 states and 85,000 census tracts (2020 data), edge weights exhibit the following distribution:

Table 1: Edge Weight Distribution Statistics (National, 2020)

Statistic	Value
Minimum weight	0.02
5th percentile	0.15
25th percentile (Q1)	0.48
Median (Q2)	1.23
75th percentile (Q3)	3.76
95th percentile	12.45
Maximum weight	48.32
Mean	2.14
Standard deviation	3.89
Coefficient of variation	1.82
<i>Skewness and heavy tails</i>	
Skewness	4.23
% weights > 10	6.2%
% weights > 20	1.4%

Interpretation:

- **Well-behaved distribution:** Median weight (1.23) is moderate, with most edges having weights 0.5-4.0
- **No pathological cases:** Maximum weight (48.32) occurs for highly similar, adjacent minority tracts in dense urban areas. This is geometrically sensible: such tracts should strongly prefer staying together
- **Positive skewness:** Right tail (high weights) reflects intentional design: demographically similar tracts receive substantially higher weights, enabling VRA compliance
- **Low weights dominate:** 75% of edges have weights < 3.76 , ensuring geographic compactness remains primary optimization objective

0.6 Regional Variation

Weight distributions vary by urban/rural geography:

Table 2: Edge Weight Statistics by Region Type

Statistic	Urban	Suburban	Rural
Median weight	1.45	1.18	0.89
95th percentile	15.23	11.87	8.34
% weights > 10	8.1%	5.9%	4.2%

Urban areas show higher weights due to:

- Greater demographic diversity (more variance in s_{demo})
- Shorter inter-tract distances (smaller $d(i, j)$)
- Concentrated minority populations (higher s_{demo} for minority-minority edges)

0.7 Sensitivity to α

To verify weights are not overly sensitive to α choice, we tested alternative scaling parameters:

Table 3: Sensitivity to Scaling Parameter α

α	Median Weight	MM Districts	Compactness (PP)
1.0	0.12	98	0.42
5.0	0.62	124	0.38
10.0 (default)	1.23	137	0.35
20.0	2.46	143	0.31
50.0	6.15	149	0.26

Results show:

- $\alpha = 10$ balances VRA compliance (137 MM districts) with compactness (0.35 Polsby-Popper)
- Doubling α to 20 yields diminishing returns (+6 MM districts) with compactness penalty
- $\alpha = 50$ produces excessive VRA focus at cost of non-compact districts

0.8 Computational Considerations

Edge weights are computed once during graph construction, before METIS partitioning. For 85,000 tracts:

- Adjacency detection: 2.3 million edges (queen contiguity)
- Weight computation: < 1 second (vectorized operations)
- Memory footprint: 37 MB (sparse edge list representation)
- METIS optimization time: 5-120 seconds depending on state size

Weights remain static during METIS refinement passes—they are not updated as partitions evolve. This ensures reproducibility: identical inputs always produce identical outputs.

0.9 Comparison to Alternative Weighting Schemes

We tested three alternative demographic similarity measures:

1. **Binary minority indicator:** $s_{demo} = 1$ if both tracts $> 50\%$ minority, else 0
2. **Hamming distance:** Count of demographic categories with $> 10\%$ difference
3. **Euclidean distance:** $1 - |\vec{v}_i - \vec{v}_j|_2 / \sqrt{2}$

Cosine similarity outperforms alternatives because:

- Binary indicator is too coarse (only captures 50% threshold, ignores gradations)
- Hamming distance requires arbitrary thresholds (10%? 20%?)
- Euclidean distance over-penalizes small differences in large categories
- Cosine similarity is threshold-free and proportionally weighted

0.10 Summary

The edge-weighting scheme $w(i, j) = \alpha \cdot s_{demo}(i, j) / (d(i, j) + \epsilon)$ provides:

- **Mathematically precise:** Clear formula with defined parameters
- **Well-behaved:** Weight distribution shows no pathological cases
- **Tunable:** Parameter α allows balancing VRA vs. compactness
- **Transparent:** All inputs (demographics, distance) are public census data
- **Reproducible:** Static weights + deterministic METIS = reproducible outputs

This specification enables independent verification and replication of our results.